

Coordinate Geometry

Covering

- Coordinates
- Drawing linear graphs
- Gradient of linear graphs
- Length and midpoint
- Equations of linear graphs
- Parallel lines
- Perpendicular lines

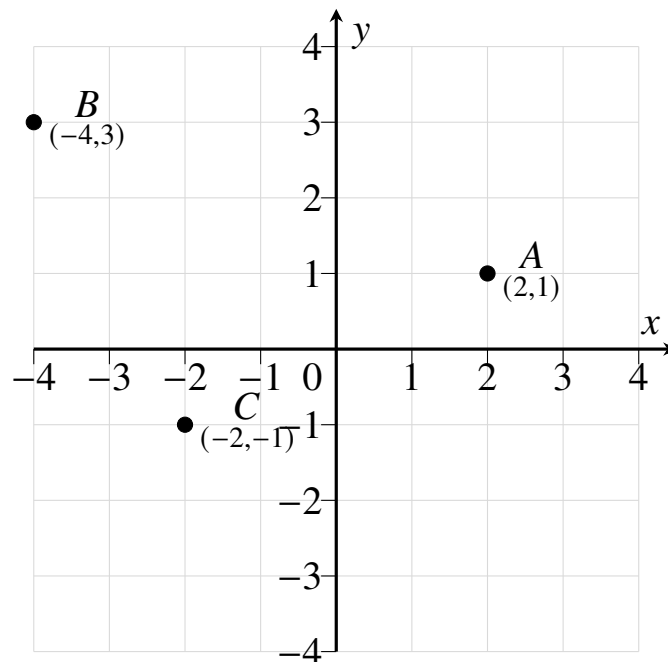
O levels and Extended

O levels and Extended

Coordinates

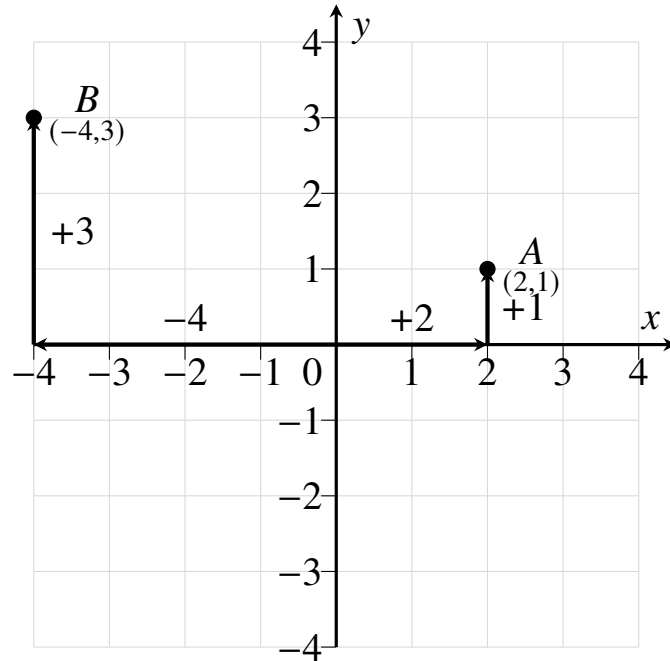
Coordinates are represented in the form (x, y) . they represent the coordinates of a point in the 2D Cartesian plane. E.g points A , B and C are on the grid below

This plane has 2 axes, x -axis and y -axis. This grid system lets us visualize geometrical objects in algebra terms.



Plotting Points

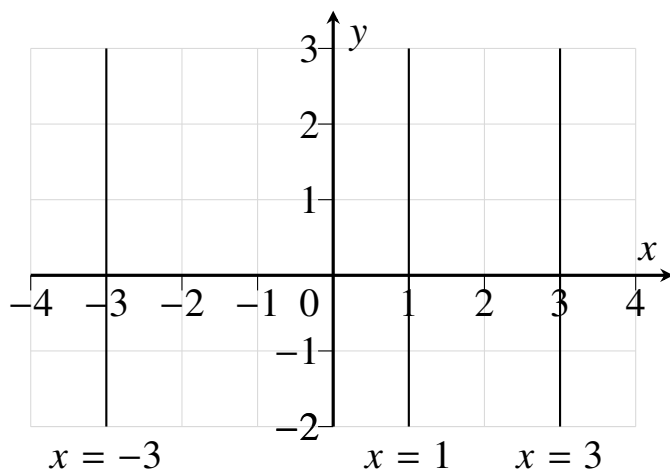
For this we first go to the x -coordinate (first value), then we move up (positive) or down (negative) to the y -coordinate. E.g let's plot the A and B points;



Equation of linear graphs

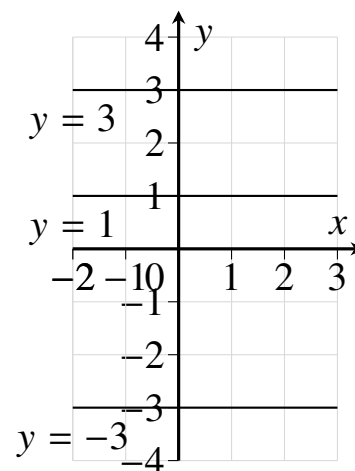
Vertical Line

For a vertical line, its equation is $x = k$, where k is the value it meets the x -axis at.
E.g these line meet at some value of x .



Horizontal Line

For a horizontal line, its equation is $y = k$, where k is the value it meets the y -axis at.
E.g these line meet at some value of y .



Sloped Line

The equation of a line(ar) graph is given by the general form $y = mx + c$, where m is the Gradient, and c is the y-intercept.

y-intercept of a line

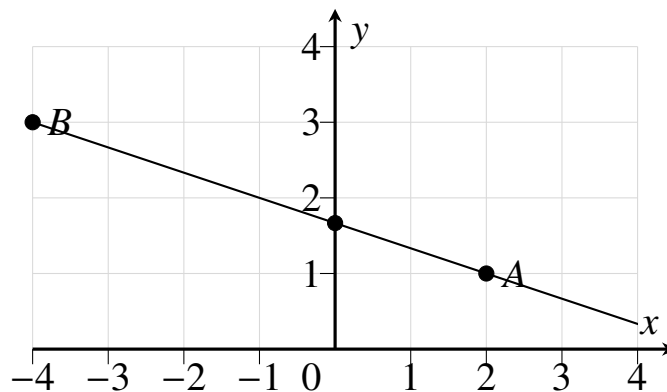
If the line is already plotted, see at what y value does it cut the y-axis. E.g we got $c = \frac{5}{3}$

If the equation of the line is given, the constant term is the y-intercept. First make sure to have x in terms of y , i.e in the form $y = mx + c$.

E.g $y = \frac{5}{3} - \frac{x}{3}$, here $\frac{5}{3}$ is y-intercept

If the equation of the line is given, but not the c , then plug in values of x and y of a point which is on the line, and solve for c . E.g we input in coordinates of point A.

$$y = c - \frac{x}{3} \rightarrow 1 = c - \frac{2}{3} \rightarrow c = \frac{5}{3}$$



Gradient of a line

If the line is already plotted, use the formula $m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\text{rise}}{\text{run}}$ such (x_1, y_1) is point 1, and (x_2, y_2) is point 2. Graph on next page shows Rise and Run, which is the displacement from point 1 to point 2. First go left or right to the x -coordinate, then up or down to the y -coordinate.

E.g for the line AB , the gradient would be, taking A and point 1 and B as point 2,

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{3 - 1}{-4 - 2} = -\frac{1}{3}. \text{ Remember to be careful of the values.}$$

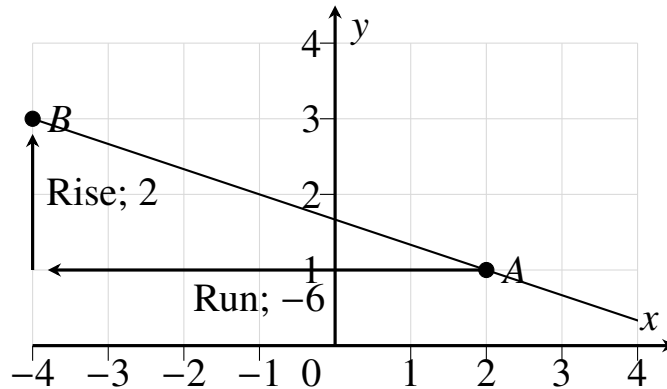
If the equation of the line is given, the x -coefficient is the gradient. First make sure to have x in terms of y , i.e in the form $y = mx + c$



Gradient of a line

If the equation of the line is given, but not the m , then plug in values of x and y of a point which is on the line, and solve for m .

$$y = xm + \frac{5}{3} \rightarrow 1 = 2m + \frac{5}{3} \rightarrow m = -\frac{1}{3}$$

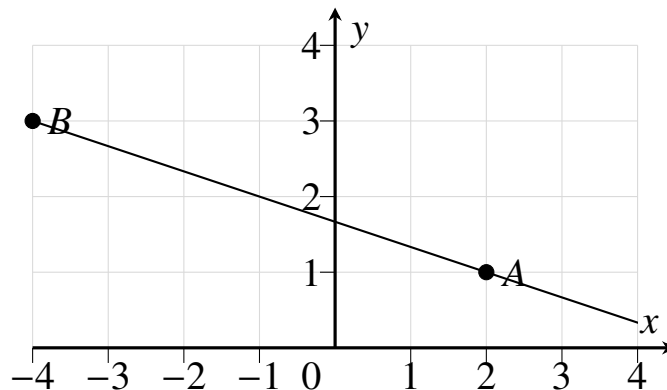


Equation of linear graphs

If we are only given 2 points to work with, we can use the formula $m(x - x_1) + y_1 = y$, where m is the gradient, (x_1, y_1) is any point which the line already or wishes to touch.

E.g we can use it to find the equation of the line for line AB .

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{3 - 1}{-4 - 2} = -\frac{1}{3} \quad m(x - x_1) + y_1 = y \rightarrow -\frac{1}{3}(x - 2) + 1 = y \rightarrow \boxed{y = \frac{x}{3} - \frac{5}{3}}$$



$m = \frac{y - y_1}{x - x_1}$, the formula taught, can be turned into the formula I suggest $m(x - x_1) + y_1 = y$

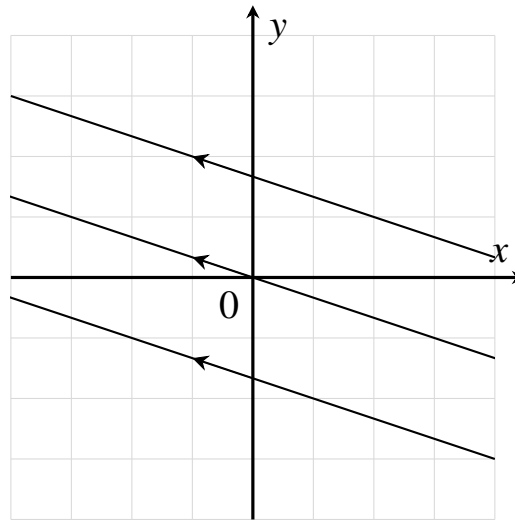
$$m = \frac{y - y_1}{x - x_1} \rightarrow m(x - x_1) = y - y_1 \rightarrow m(x - x_1) + y_1 = y$$



Parallel Line

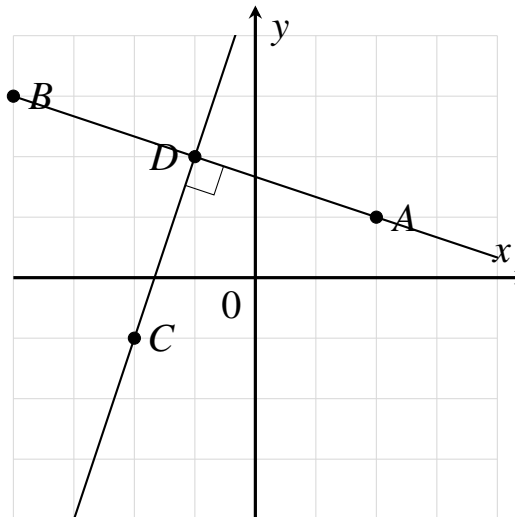
A parallel line is whose Gradient is the same, but not the y-intercept. They never meet the line "parallel" to them.

E.g the three lines are "parallel" because they never meet ever and their gradient is the same.



Perpendicular Lines

A perpendicular line makes a 90° with the "original" line. E.g Line CD is perpendicular to Line AB because they make a 90° at their intersection.



To find the gradient of the perpendicular line, m_2 , we use the formula $m_2 = -\frac{1}{m_1}$, where m_1 is the "original" line's gradient.

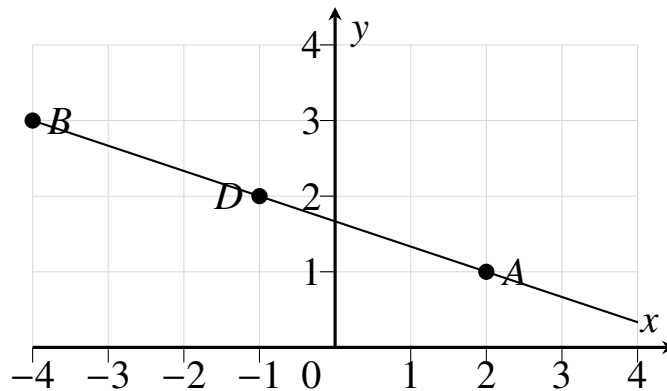


Mid point of two points**O levels and Extended**

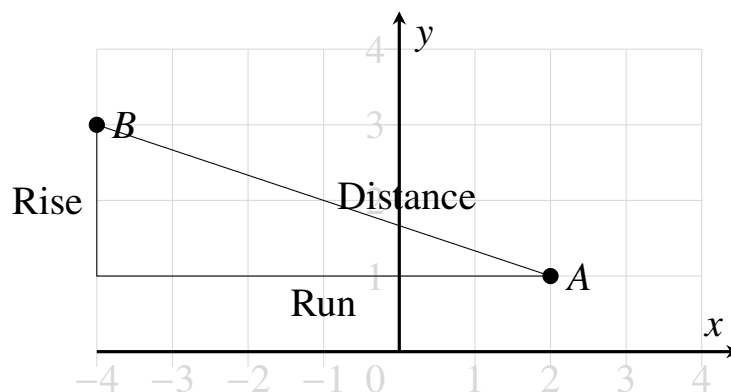
To find the "Mid point", or the point which is halfway between 2 points, we take the average of the coordinates, or use the formula $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$.

E.g to find the midpoint of A and B , D we do:

$$D = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right) = \left(\frac{2 + (-4)}{2}, \frac{1 + 3}{2}\right) = \left(\frac{-2}{2}, \frac{4}{2}\right) = (-1, 2)$$

**Length of a line/Distance between points****O levels and Extended**

For this we use the formula $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$, here $y_2 - y_1$ and $x_2 - x_1$ indeed are familiar, them being Rise and Run, but here they are the legs of a triangle for which the hypotenuse is the distance we want to calculate. On the graph we can see this triangle.

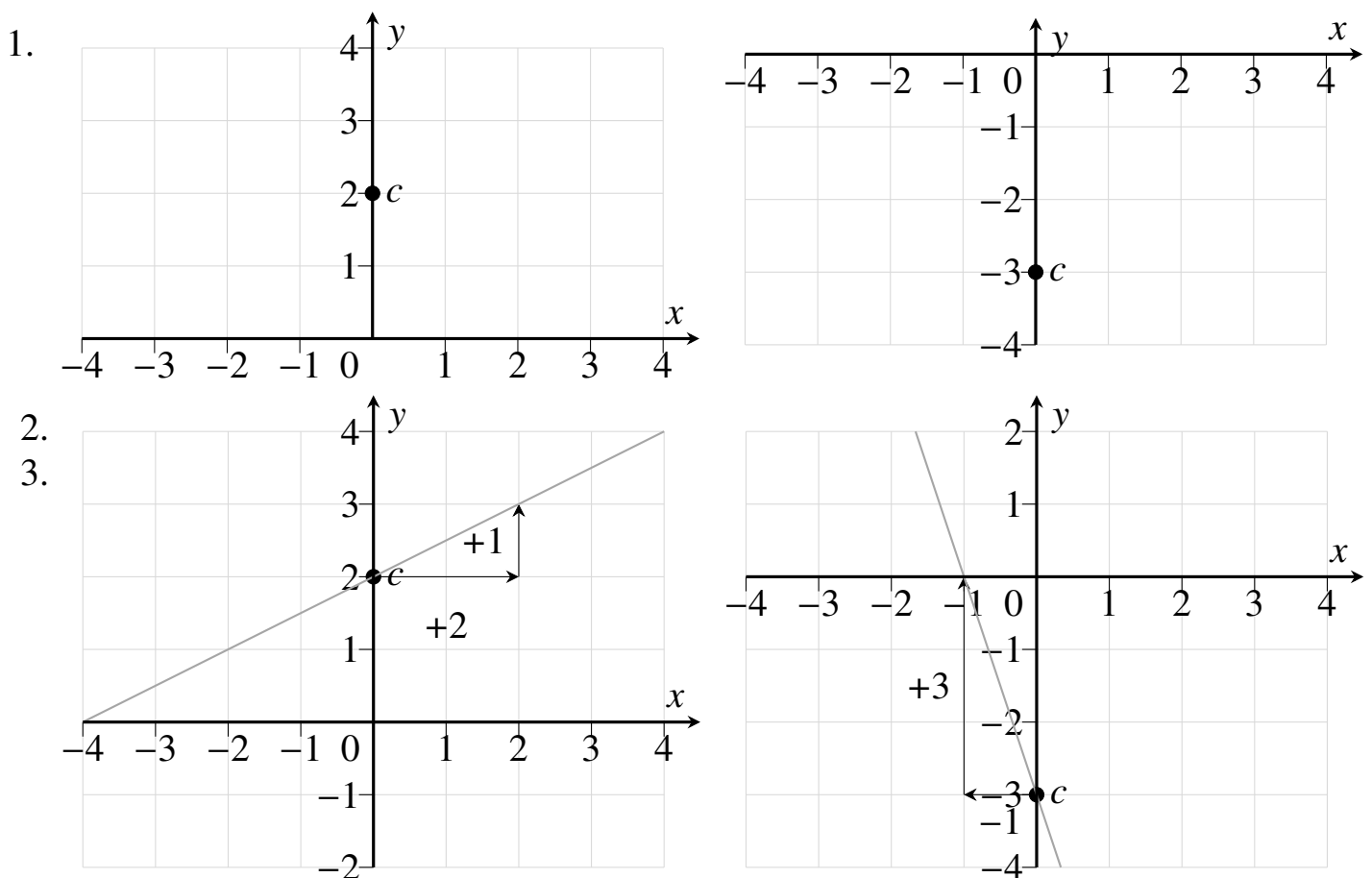


Drawing linear graphs

To draw, we first make the equation in terms of y if needed.

1. Next we first go to the y -intercept, $(0, c)$, if there is no constant term, y -intercept is 0.
2. Next for the gradient. These start from the intercept point.
 - i. If its a whole number, in the x go 1 unit to the left(negative) or right(positive), depending on the sign. Then in the y go the unit of the value (ignore sign).
 - ii. If its a fraction, in the x go the unit of the denominator to the left(negative) or right(positive), depending on the sign. Then in the y go the unit of the numerator (ignore sign).
3. Now using your HB pencil and a straight edge(ruler), connect the 2 points and continue the line to the edges of the graph on the paper.

E.g here we see plotting $y = \frac{1}{2}x + 2$ and $y = -3x - 3$



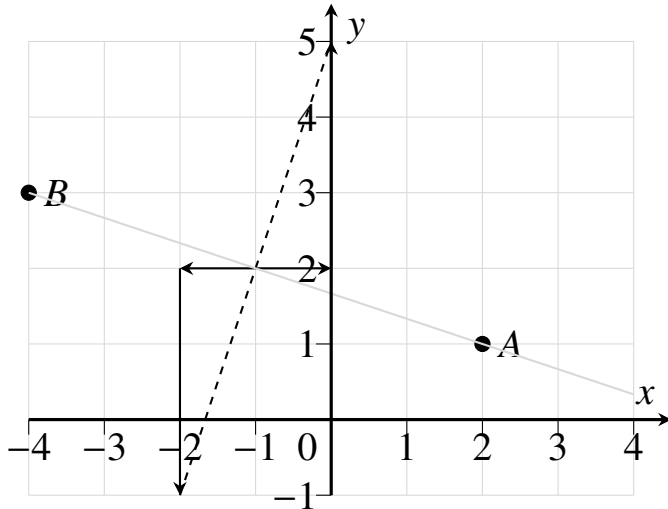
Perpendicular Bisector

O levels and Extended

Perpendicular bisector of a line, say AB , is a line which passes through the midpoint of the 2 points. It is found by using 1. $m(x - x_1) + y_1 = y$ where m is the perpendicular gradient, $m_2 = -\frac{1}{m_1}$ and 2. (x_1, y_2) is the mid point.

E.g Perpendicular Bisector of line AB , is:

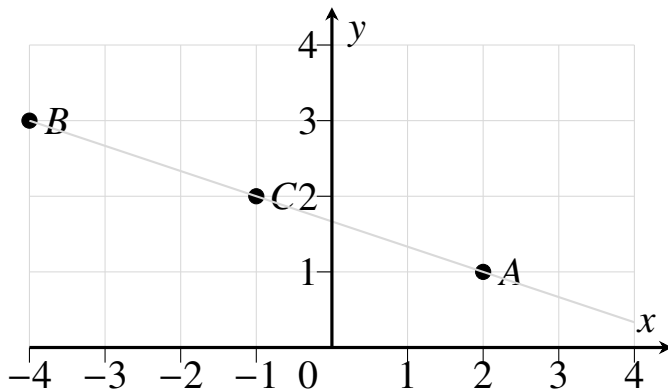
1.



$$m_{AB} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{3 - 1}{-4 - 2} = -\frac{1}{3}$$

$$m_2 = -\frac{1}{m_1} = -\frac{1}{-\frac{1}{3}} = 3$$

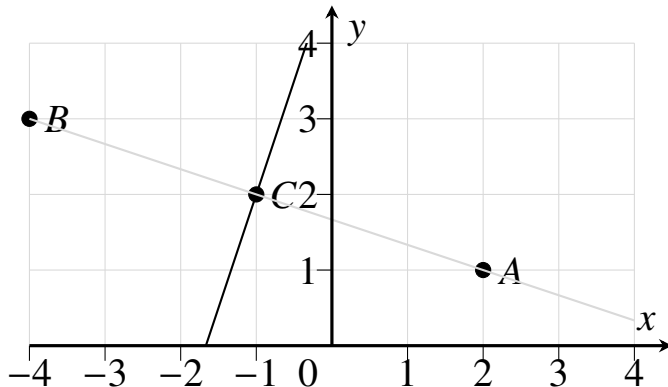
2.



$$C = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$= \left(\frac{2 + (-4)}{2}, \frac{1 + 3}{2} \right) = (-1, 2)$$

3.



$$y = m(x - x_1) + y_1$$

$$= 3(x - (-1)) + 2 = 3x + 5 = y$$

